Lesson 2: Spreadsheet Automation: A Dashboard That Updates Itself

Module 1 | 60 minutes | Formulas that update | Adults (18+) · ages 18–65

Objective: Get AI to write formulas with several conditions (SUMIFS, COUNTIFS, IF, XLOOKUP, TEXT) and conditional formatting rules, clean a messy sheet with formulas, build a weekly dashboard that updates when the date or data changes, and verify every number.

Success criteria

- I can get formulas with several conditions (SUMIFS, COUNTIFS) from a plain-words request.
- I can clean a messy sheet with TRIM and a check column.
- I can build a dashboard that updates when the date or data changes, with conditional formatting.
- I can verify a dashboard number with a filter and explain any difference.

Key vocabulary

- **SUMIFS / COUNTIFS**: Add up or count only the rows that match several conditions at once.
- **TRIM**: Removes extra spaces at the start or end of text, so “Baner ” matches “Baner”.
- **TEXT function**: Shows a number or date in a neat format, like 15,310 or 19 Oct 2026.
- **conditional formatting**: A rule that colours cells automatically, like red when sales are below target.
- **dashboard**: One page of key numbers that updates from your data.
- **check column**: An extra column with a formula that shows OK or Check for each row.

You will need

- Projector or screen share with Kitabash and the AI tools for this lesson
- Devices: 1 per learner (laptop, tablet or phone)
- Learner accounts only where allowed: see the course safety notes
- Printed worksheet (optional)
- Project files: weekly_orders.csv, l02_dashboard_kit.md, l02_teacher_demo_and_answer_key.md

Timed lesson plan (60 min)

Time	Stage	What happens
5 min	**Warm-up game**	“Human SUMIFS” stand-up game (see Unplugged).
4 min	**Watch + poll**	Play the lesson video. Fist-to-five poll: “How confident are you with SUMIF from Level 1?”

8 min	**Live demo**	Problems and fixes, the check column, the dashboard on one date cell, the one-day-off catch, colour rules, the date-change test.
15 min	**Build along (pairs)**	Build steps 1–6 in pairs. Halfway check: “Type your B3 in the chat.” Everyone should reach ₹15,310 before adding colour.
8 min	**Prompt Labs**	Prompt Lab: a formula with several conditions (the credit cell), then the check column and colour rule. Paste the AI’s answers into the dashboard.
6 min	**Team race**	Quick competition: the function match and dashboard build order, two teams, fastest correct wins. Then “Which number do you trust?” as a class poll.
5 min	**Show-and-tell**	Two learners share screens: change B1 to 12 Oct live and show everything updating and turning red.
5 min	**Probe**	New situation: the dashboard that shows zero (individual).
4 min	**Wrap-up**	Exit ticket; homework: add one number your own work needs.

Unplugged warm-up

Human SUMIFS (5 min): give 10 volunteers order cards (date, status, amount; one card dated next Monday, one cancelled). Call out conditions one at a time: “Stand up if your order is in the week of 19 Oct... stay standing only if Delivered... only if paid on Credit.” The class adds the standing amounts. Then ask: “Who wrongly stood up at the first call?” (the next-Monday card). That is exactly the mistake we will catch in a formula today.

Teaching the idea

Level 1 taught SUMIF, XLOOKUP and checking totals on a one-off report. Today the sheet does the work every week. Three ideas: formulas with several conditions (SUMIFS, COUNTIFS), cleaning with formulas (TRIM and a check column), and a dashboard that hangs off one date cell so it updates itself. Conditional formatting adds warning colours. The caught mistake is an off-by-one date: "<=" B1+7 quietly counts the next Monday. Learners prove their dashboard with a filter check before trusting it, and test it by changing the date.

Analogy: A self-updating dashboard is like a car’s fuel gauge: you do not measure the tank every morning, the gauge updates itself, but you still check it is accurate after any repair.

Live demo: 1) Open weekly_orders.csv in Google Sheets as the Orders tab. 2) Upload it to Claude or ChatGPT with the “list problems and fixes, do not change the file” prompt. 3) Add the check column =IF(H2=F2*G2,"OK","Check") and fill down: three rows say Check (the two text amounts and the blank). Ask learners why the duplicate does NOT show: it adds up, it is just there twice. Fix all four problems live. 4) Make the Dashboard tab: B1 19-Oct-2026, the TEXT label in A2. 5) Ask Gemini for “delivered sales in that week” without saying where the week ends; if it uses "<=", show the ₹16,530 and the filter check ₹15,310. If it gets it right, type the "<=" version yourself to show the trap. 6) Add B4–B8. 7) Add the red/green rule on B3 and change B1 to 12 Oct (₹14,090, red). 8) Put =TODAY()-WEEKDAY(TODAY(),3) in B1 for a moment, then back to 19 Oct.

Questions to ask

Ask	Listen for
What does SUMIFS do that SUMIF cannot?	It checks several conditions at once, like the week, the status and the payment type.
Why does every formula point at B1?	So changing one date updates the whole dashboard.
Why use "<" B1+7 and not "<=" B1+7?	"<=" also includes the next Monday, so the week would have eight days.
How do you verify the dashboard?	Filter the data for the same week and condition and compare the Sum with the dashboard number.
Why clean before building formulas?	Text amounts, blanks, spaces and duplicates make SUMIFS give wrong totals without any error message.

Differentiation

Support
Give a half-built Dashboard with B1, A2 and B3 filled in; learners ask the AI for B4–B8 by copying and changing the B3 prompt. Pair spreadsheet beginners with confident learners.

Stretch
Add a “sales by item” table with SUMIFS per item, a “top item this week” cell (ask the AI for INDEX/MATCH with MAX, or SORTBY), and an XLOOKUP rate check against a small price list tab.

Common misconceptions

Misconception	How to fix it
If a formula gives no error, it is correct.	Wrong dates or missing conditions give wrong numbers with no error. Verify with a filter.
A typed date is always a real date.	Sometimes it is stored as text. Re-enter it or use DATE(), then test.
Self-updating means I never check it again.	Check one number by hand every week, and after any change to the sheet.
I must learn every function by heart.	Describe the columns and conditions; the AI writes and explains the formula.

Assessment

Use the Prompt Lab results (auto-checked), the quick checks and the Probe answers. Written answers arrive in your review queue.

Exit ticket
Your dashboard and your filter check differ by the value of exactly two orders on one date. What do you check first?

Homework
Finish the dashboard and add one number your own work needs (for example “orders still pending”). Optional: build the same dashboard on an anonymised copy of a real sheet from your work.

Answer key — Kitabash activities

1. Prompt Lab: a formula with several conditions (perform · ai lab)

Checklist: Name your app · Give the columns · Point at the week start cell · Include the Delivered condition · Include the Credit condition · Say exactly where the week ends · Ask for an explanation

Strong example:

```
In Google Sheets, my Orders tab has Order Date in column A, Amount in column H, Payment in column I and Status in column J. On my Dashboard tab, B1 holds the Monday the week starts. Write a SUMIFS formula for the total Amount where the date is from B1 up to Sunday (before the next Monday), Status is Delivered and Payment is Credit. Explain it in simple words.
```

2. Prompt Lab: a check column and a colour rule (perform · ai lab)

Checklist: Name your app · Give the columns · Ask for an OK / Check formula · Say the rule (quantity times rate) · Ask for conditional formatting · Ask where to click

Strong example:

```
In Google Sheets, my Orders tab has Qty in column F, Rate in column G and Amount in column H. (1) Give me a formula for column K that shows “OK” when Amount equals Qty times Rate and “Check” when it does not. (2) Give me a conditional formatting rule that colours the whole row red when column K says Check. Tell me the steps and which menu to use.
```

3. Match each tool to the job it does on your dashboard (perform · matching)

SUMIFS → Add amounts that match several conditions, like this week and delivered · COUNTIFS → Count orders that match several conditions · TRIM → Remove extra spaces, like “Baner ” with a space at the end · TEXT → Show a date or amount as a neat label, like 19 Oct 2026 · XLOOKUP → Find an item’s price in a separate price list · Conditional formatting → Colour a cell red when sales are below target

4. Dashboard build order (perform · ordering)

Order: Import the data into an Orders tab → Clean it: duplicates, spaces, text amounts, blanks → Put the week start date in one cell → Ask AI for formulas that point at that cell → Add conditional formatting for targets → Verify one number by hand with a filter → Change the date to test that everything updates

5. Which number do you trust? (perform · multiple choice)

Answer: The formula uses "<=" B1+7, so it also counts orders on Monday 26 Oct

6. New situation: the dashboard that shows zero (probe · multiple choice)

Answer: B1 was stored as text, not a real date, so the date comparisons find nothing; re-enter it as a date (for example 02-Nov-2026 or =DATE(2026,11,2))

Answer key — worksheet

1. **Key words:** SUMIFS / COUNTIFS = Add up or count only the rows that match several conditions at once. · TRIM = Removes extra spaces at the start or end of text, so “Baner ” matches “Baner”. · TEXT function = Shows a number or date in a neat format, like 15,310 or 19 Oct 2026. · conditional formatting = A rule that colours cells automatically, like red when sales are below target.
2. **Prompt Lab: a formula with several conditions:** Include: Name your app; Give the columns; Point at the week start cell; Include the Delivered condition; Include the Credit condition; Say exactly where the week ends; Ask for an explanation. Example: In Google Sheets, my Orders tab has Order Date in column A, Amount in column H, Payment in column I and Status in column J. On my Dashboard tab, B1 holds the Monday the week starts. Write a SUMIFS formula for the total Amount where the date is from B1 up to Sunday (before the next Monday), Status is Delivered and Payment is Credit. Explain it in simple words.
3. **Prompt Lab: a check column and a colour rule:** Include: Name your app; Give the columns; Ask for an OK / Check formula; Say the rule (quantity times rate); Ask for conditional formatting; Ask where to click. Example: In Google Sheets, my Orders tab has Qty in column F, Rate in column G and Amount in column H. (1) Give me a formula for column K that shows “OK” when Amount equals Qty times Rate and “Check” when it does not. (2) Give me a conditional formatting rule that colours the whole row red when column K says Check. Tell me the steps and which menu to use.
4. **Match each tool to the job it does on your dashboard:** SUMIFS → Add amounts that match several conditions, like this week and delivered · COUNTIFS → Count orders that match several conditions · TRIM → Remove extra spaces, like “Baner ” with a space at the end · TEXT → Show a date or amount as a neat label, like 19 Oct 2026 · XLOOKUP → Find an item’s price in a separate price list · Conditional formatting → Colour a cell red when sales are below target
5. **Dashboard build order:** 1 Import the data into an Orders tab · 2 Clean it: duplicates, spaces, text amounts, blanks · 3 Put the week start date in one cell · 4 Ask AI for formulas that point at that cell · 5 Add conditional formatting for targets · 6 Verify one number by hand with a filter · 7 Change the date to test that everything updates

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